

MODUL

Pertemuan-14: API dan Android

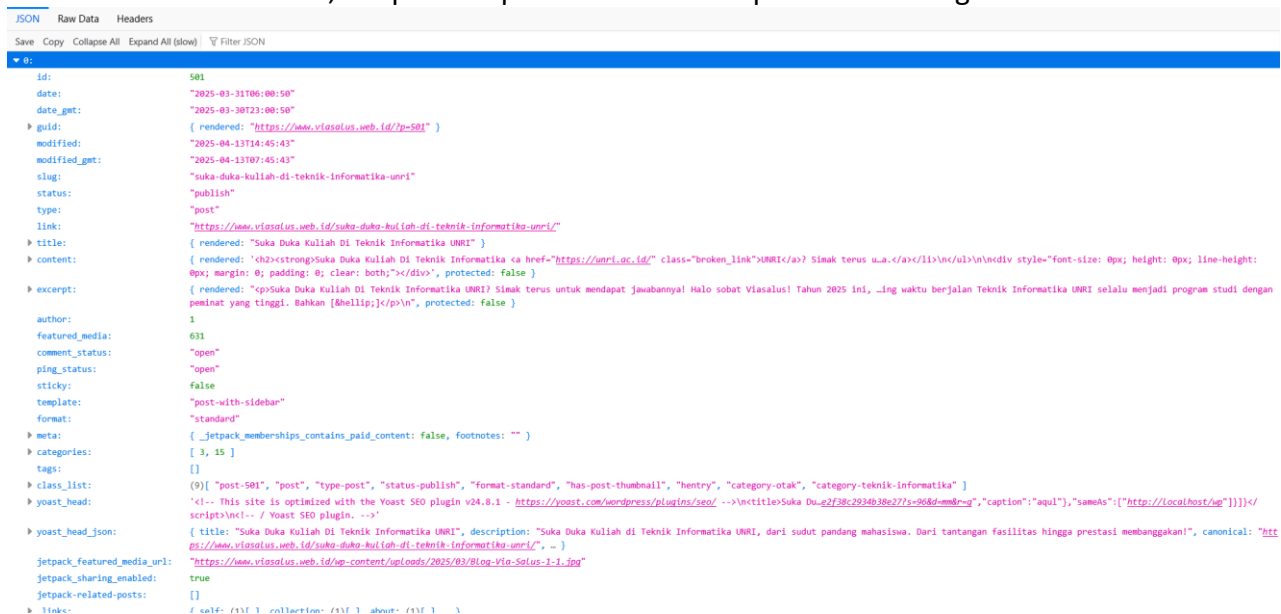
A. Capaian Kompetensi

Capaian kompetensi dalam pembahasan ini adalah sebagai berikut:

- 1) Mahasiswa mampu membuat sebuah alur activity yang terhubung dan mengirimkan sebuah nilai dari activity ke activity menggunakan intent.
- 2) Mahasiswa mampu menggunakan komponen dasar penyusun antar muka android seperti button, textview dan toast
- 3) Mahasiswa mampu memahami, menjelaskan dan mempraktekkan penggunaan fragment
- 4) Mahasiswa mampu dalam mengintegrasikan API eksternal ke dalam aplikasi android

B. Persiapan

- 1) Buatlah sebuah API dari sebuah website yang telah dibuat, pastikan API tersebut dapat diakses!, dalam modul ini kita gunakan contoh API dari link berikut: <https://www.viasalus.web.id/wp-json/wp/v2/posts>
- 2) Pastikan API bisa diakses, adapun tampilan API dari tahap 1 adalah sebagai berikut:



```
JSON Raw Data Headers
Save Copy Collapse All Expand All (slow) Filter JSON
▼ 0:
id: 501
date: "2025-03-31T06:00:50"
date_gmt: "2025-03-30T23:00:50"
guid: { rendered: "https://www.viasalus.web.id/?p=501" }
modified: "2025-04-13T14:45:43"
modified_gmt: "2025-04-13T07:45:43"
slug: "suka-duka-kuliah-di-teknik-informatika-unri"
status: "publish"
type: "post"
link: "https://www.viasalus.web.id/suka-duka-kuliah-di-teknik-informatika-unri/"
title: { rendered: "Suka Duka Kuliah Di Teknik Informatika UNRI" }
content: { rendered: "ch2<strong>Suka Duka Kuliah Di Teknik Informatika ca href="https://unri.ac.id/" class="broken_link"HNRIC/a? Simak terus u.a.</a></li><ul><li><div style="font-size: 0px; height: 0px; line-height: 0px; margin: 0; padding: 0; clear: both;"</div>, protected: false }
excerpt: { rendered: "cpSuka Duka Kuliah Di Teknik Informatika UNRI? Simak terus untuk mendapat jawabannya! Halo sobat Viasalus! Tahun 2025 ini, _ing waktu berjalan Teknik Informatika UNRI selalu menjadi program studi dengan peminat yang tinggi. Bahkan [hellip];</p>\n", protected: false }
author: 1
featured_media: 631
comment_status: "open"
ping_status: "open"
sticky: false
template: "post-with-sidebar"
format: "standard"
meta: { _jetpack_memberships_contains_paid_content: false, footnotes: "" }
categories: [ 3, 15 ]
tags: []
class_list: (9) ["post-501", "post", "type-post", "status-publish", "format-standard", "has-post-thumbnail", "hentry", "category-otak", "category-teknik-informatika"]
yoast_head: '<!-- This site is optimized with the Yoast SEO plugin v24.8.1 - https://yoast.com/wordpress/plugins/seo/ -->\n<title>Suka Du-e2f38c2934b38e2773-968d-m8r-g", "caption": "aqui", "size": ["http://localhost/wp"]}]</script>\n<!-- / Yoast SEO plugin, -->'
yoast_head_json: {
  title: "Suka Duka Kuliah Di Teknik Informatika UNRI",
  description: "Suka Duka Kuliah di Teknik Informatika UNRI, dari sudut pandang mahasiswa. Dari tantangan fasilitas hingga prestasi membanggakan!",
  canonical: "https://www.viasalus.web.id/wp-content/uploads/2025/03/8log-Via-Salus-1-1.jpg"
  jetpack_featured_media_url:
  jetpack_sharing_enabled: true
  jetpack-related-posts: []
}
links: [ self: /?f=3, collection: /?f=3, about: /?f=3, ... ]
```

Nb. Format output API yang berbeda dari akan membuat code tidak bisa melakukan parse object dengan sempurna.

C. Mengintegrasikan API dengan Aplikasi

- 1) Buat sebuah project dengan nama wpapiapp!
- 2) Tambahkan dependency volley, glide, dan glide compiler pada file gradle (Gradle Scripts→build.gradle.kts), sehingga menjadi seperti berikut:

```
...
dependencies {
    implementation("androidx.appcompat:appcompat:1.7.0")
    implementation("com.google.android.material:material:1.12.0")
    implementation("androidx.constraintlayout:constraintlayout:2.2.1")
    testImplementation("junit:junit:4.13.2")
}
```

```

    androidTestImplementation("androidx.test.ext:junit:1.2.1")
    androidTestImplementation("androidx.test.espresso:espresso-core:3.6.1")
    implementation("com.android.volley:volley:1.2.1")
    implementation("com.github.bumptech.glide:glide:4.16.0")
    annotationProcessor("com.github.bumptech.glide:compiler:4.16.0")
}

```

3) Buat class java dengan nama: PostModel, PostAdapter, GlideImageGetter, dan CustomDrawablePlaceHolder.

4) Pada file PostModel, ketikkan syntax berikut:

```

package com.example.wpapiapp;

public class PostModel {
    String title, content, imageUrl;

    public PostModel(String title, String content, String imageUrl) {
        this.title = title;
        this.content = content;
        this.imageUrl = imageUrl;
    }
}

```

5) Pada file PostAdapter, ketikkan syntax berikut:

```

package com.example.wpapiapp;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
import com.bumptech.glide.Glide;

import java.util.List;

public class PostAdapter extends RecyclerView.Adapter<PostAdapter.ViewHolder> {

    Context context;
    List<PostModel> posts;
    OnPostClickListener listener;

    interface OnPostClickListener {
        void onClick(PostModel post);
    }

    public PostAdapter(Context context, List<PostModel> posts,
        OnPostClickListener listener) {
        this.context = context;
        this.posts = posts;
        this.listener = listener;
    }

    @NonNull
    @Override
    public PostAdapter.ViewHolder onCreateViewHolder(@NonNull ViewGroup parent,
        int viewType) {
        return new

```

```

ViewHolder(LayoutInflater.from(context).inflate(R.layout.item_post, parent,
false));
    }

    @Override
    public void onBindViewHolder(@NonNull PostAdapter.ViewHolder holder, int
position) {
        PostModel post = posts.get(position);
        holder.title.setText(post.title);
        Glide.with(context).load(post.imageUrl).into(holder.image);
        holder.itemView.setOnClickListener(v -> listener.onClick(post));
    }

    @Override
    public int getItemCount() {
        return posts.size();
    }

    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView title;
        ImageView image;

        public ViewHolder(@NonNull View itemView) {
            super(itemView);
            title = itemView.findViewById(R.id.postTitle);
            image = itemView.findViewById(R.id.postImage);
        }
    }
}

```

6) Pada file GlideImageGetter, ketikkan syntax berikut:

```

package com.example.wpapiapp;

import android.content.Context;
import android.graphics.drawable.Drawable;
import android.text.Html;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;

import com.bumptech.glide.Glide;
import com.bumptech.glide.request.target.CustomTarget;
import com.bumptech.glide.request.transition.Transition;

public class GlideImageGetter implements Html.ImageGetter {
    private final TextView textView;
    private final Context context;

    public GlideImageGetter(TextView textView, Context context) {
        this.textView = textView;
        this.context = context;
    }

    @Override
    public Drawable getDrawable(String source) {
        CustomDrawablePlaceholder drawable = new CustomDrawablePlaceholder();
        Glide.with(context)
            .load(source)
            .into(new CustomTarget<Drawable>() {

```

```

        @Override
        public void onResourceReady(@NonNull Drawable resource,
@Nullable Transition<? super Drawable> transition) {
            int width = textView.getWidth();
            int imgWidth = resource.getIntrinsicWidth();
            int imgHeight = resource.getIntrinsicHeight();
            float scale = (float) width / imgWidth;
            int scaledHeight = (int) (imgHeight * scale);
            resource.setBounds(0, 0, width, scaledHeight);
            drawable.setBounds(0, 0, width, scaledHeight);
            drawable.setDrawable(resource);
            textView.setText(textView.getText()); // redraw
        }

        @Override
        public void onLoadCleared(@Nullable Drawable placeholder)
    {}

    });
    return drawable;
}
}

```

7) Pada file CustomDrawablePlaceHolder, ketikkan syntax berikut:

```

package com.example.wpapiapp;

import android.graphics.Canvas;
import android.graphics.drawable.Drawable;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;

public class CustomDrawablePlaceHolder extends Drawable {
    private Drawable drawable;

    @Override
    public void draw(@NonNull Canvas canvas) {
        if (drawable != null) drawable.draw(canvas);
    }

    public void setDrawable(Drawable drawable) {
        this.drawable = drawable;
    }

    @Override public void setAlpha(int alpha) {}
    @Override public void setColorFilter(@Nullable android.graphics.ColorFilter
colorFilter) {}
    @Override public int getOpacity() { return
android.graphics.PixelFormat.TRANSPARENT; }
}

```

8) Selanjutnya buat activity baru dengan nama PostDetailActivity.

9) Pada file activity_main.xml, ubah menjadi seperti berikut:

```

<androidx.recyclerview.widget.RecyclerView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>

```

10) Pada file activity_post_detail, ubah menjadi seperti berikut:

```

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"

```

```

        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="16dp">
        <ImageView
            android:id="@+id/imageDetail"
            android:layout_width="match_parent"
            android:layout_height="200dp"
            android:scaleType="centerCrop" />
        <TextView
            android:id="@+id/titleDetail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:textStyle="bold"
            android:layout_marginTop="10dp" />
        <androidx.core.widget.NestedScrollView
            android:layout_width="match_parent"
            android:layout_height="0dp"
            android:layout_weight="1">
            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical">
                <TextView
                    android:id="@+id/contentDetail"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:layout_marginTop="8dp" />
            </LinearLayout>
        </androidx.core.widget.NestedScrollView>
    </LinearLayout>

```

- 11) Selanjutnya buat file pada folder layout dengan nama item_post.xml, dan isikan sesuai syntax berikut!

```

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="wrap_content"
    android:orientation="vertical" android:padding="8dp">
    <ImageView
        android:id="@+id/postImage"
        android:layout_width="match_parent"
        android:layout_height="200dp" android:scaleType="centerCrop"/>
    <TextView
        android:id="@+id/postTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" android:textStyle="bold"
        android:padding="8dp"/>
</LinearLayout>

```

- 12) Pada file PostDetailActivity, ubah code menjadi seperti berikut:

```

package com.example.wpapiapp;

import android.graphics.drawable.Drawable;
import android.os.Bundle;
import android.text.Html;
import android.text.Spanned;
import android.text.method.LinkMovementMethod;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

```

```

import com.bumptech.glide.Glide;

public class PostDetailActivity extends AppCompatActivity {

    TextView contentText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_post_detail);

        TextView title = findViewById(R.id.titleDetail);
        contentText = findViewById(R.id.contentDetail);
        ImageView image = findViewById(R.id.imageDetail);

        String postTitle = getIntent().getStringExtra("title");
        String htmlContent = getIntent().getStringExtra("content");
        String imageUrl = getIntent().getStringExtra("image");

        title.setText(postTitle);
        Glide.with(this).load(imageUrl).into(image);

        contentText.setMovementMethod(LinkMovementMethod.getInstance());
        Spanned htmlSpanned = Html.fromHtml(htmlContent,
        Html.FROM_HTML_MODE_LEGACY, new GlideImageGetter(contentText, this), null);
        contentText.setText(htmlSpanned);
    }
}

```

13) Pada file MainActivity, ubah code menjadi seperti berikut:

```

package com.example.wpapiapp;

import android.content.Intent;
import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import com.android.volley.Request;
import com.android.volley.toolbox.JsonArrayRequest;
import com.android.volley.toolbox.Volley;
import org.json.JSONArray;
import org.json.JSONObject;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    RecyclerView recyclerView;
    ArrayList<PostModel> postList = new ArrayList<>();
    PostAdapter adapter;
    final String API_URL = "https://www.viasalus.web.id/wp-json/wp/v2/posts";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        recyclerView = findViewById(R.id.recyclerView);
        adapter = new PostAdapter(this, postList, post -> {

```

```

        Intent intent = new Intent(this, PostDetailActivity.class);
        intent.putExtra("title", post.title);
        intent.putExtra("content", post.content);
        intent.putExtra("image", post.imageUrl);
        startActivity(intent);
    });
    recyclerView.setLayoutManager(new LinearLayoutManager(this));
    recyclerView.setAdapter(adapter);
    fetchPosts();
}

void fetchPosts() {
    JSONArrayRequest request = new JSONArrayRequest(Request.Method.GET,
API_URL, null,
        response -> {
            try {
                for (int i = 0; i < response.length(); i++) {
                    JSONObject obj = response.getJSONObject(i);
                    String title =
obj.getJSONObject("title").getString("rendered");
                    String content =
obj.getJSONObject("content").getString("rendered");
                    String imageUrl =
obj.optString("jetpack_featured_media_url");
                    postList.add(new PostModel(title, content,
imageUrl));
                }
                adapter.notifyDataSetChanged();
            } catch (Exception e) {
                Toast.makeText(this, "Parsing error",
Toast.LENGTH_SHORT).show();
            }
        },
        error -> Toast.makeText(this, "Fetch error",
Toast.LENGTH_SHORT).show()
    );
    Volley.newRequestQueue(this).add(request);
}
}

```

14) Jalankan Aplikasi!

D. Tugas

- 1) Pahami Setiap tahapan dari Aktivitas Perkuliahan (Bagian C), lakukan ekspolarasi library/package untuk melakukan parse API dari berbagai sumber!